Active Subgraph Skip Profiles: Layer-level Computation Routing in GPT-2 Family Models

graphs_0505 analysis — reasoning suite

LLM Path Analysis Project

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Slide 1 — Experiment Setup

Models

Model	Layers	Heads	Params
`gpt2`	12	12	117M
`gpt2-medium`	24	16	345M

Both run on CPU, `eval()` mode.

Task suite: `reasoning` (6 tasks)

Label	Prompt (truncated)
1-hop chain	*Alice is the mother of Bob. Alice's chi*
2-hop chain	*Alice → Bob → Carol. Alice's grandch*
3-hop chain	*Alice → Bob → Carol → Dana. . . . gr*
Cat. syllogism	*All birds have wings. Penguins are birc*
Negation+ded.	*No reptiles are warm-blooded. All man*
Counterfactual	*In a world where cats bark. . . barking i*

Active/inactive criterion (new)

Each block (attention or FFN) is classified by its **output-to-input magnitude ratio**:

$$r^{(l)}_{\text{attn}} = \frac{\|\texttt{attn\_out}^{(l)}\|_2}{\|\texttt{resid\_pre}^{(l)}\|_2}, \qquad r^{(l)}_{\text{ffn}} = \frac{\|\texttt{mlp\_out}^{(l)}\|_2}{\|\texttt{resid\_mid}^{(l)}\|_2}$$

Inactive (skip-dominant): $r < 0.05$ ⇒ thick teal U-arc
Active (block transforms): $r \geq 0.05$ ⇒ thin dashed arc

Attribution scoring (AtP)

Per-head score at `hook_z`:

$$s(l, h) = \text{mean}_{t,d} \,|\, \nabla_z \log p \odot z \,|$$

Nucleus coverage $\rho = 0.90$: keep fewest edges covering 90% of total attribution mass.
Model params frozen (`requires_grad_(False)`) before backward to avoid $O(P)$ gradient cost.

Slide 2 — Finding 1: Opposing First-Layer Strategies

GPT-2 — attention-dominant entry

Layer	Attn ratio r		FFN ratio r	
	status	value	status	value
L0	✓	**5.2–5.9**	✓	**1.2–1.8**
L1	✓	0.17	✓	0.43–0.71
L2	✓	0.08–0.11	✓	1.12–1.84
L3	varies	0.03–0.07	✓	0.11–0.13

L0 attention $r \approx 5.5$: attn output is $5\times$ the stream magnitude — heads dominate the very first transformation.

L2 FFN is a secondary spike ($r \approx 1.5$), the first "feature extraction" checkpoint.

GPT-2-medium — FFN-dominant entry

Layer	Attn ratio r		FFN ratio r	
	status	value	status	value
L0	✓	0.34–0.36	✓	**36–38**
L1	✓	0.14–0.16	✓	0.19–0.21
L2	✓	0.11–0.15	✓	0.19–0.28
L3	✓	0.06–0.07	✓	0.86–1.71

L0 FFN $r \approx 37$: FFN output is $37\times$ the stream magnitude — the first feed-forward layer essentially rewrites the token.

Attention at L0 is present ($r \approx 0.35$) but plays a minor role compared to the FFN.

Interpretation

Both models always activate L0, but which *sub-block* dominates differs entirely. GPT-2 is **attention-first**; GPT-2-medium is **FFN-first**. This suggests the two training runs converge on different strategies for converting token embeddings into distributional representations — a difference invisible to loss curves but visible in magnitude profiles.

Slide 3 — Finding 2: Attention Silence Zones and the Bilateral Dead Zone

Attention-silent layers per task

Task	GPT-2 attn × layers	GPT-2-med attn × layers
1-hop chain	L3–L9 (7 layers)	L4–L20 (17 layers)
2-hop chain	L3–L4 (2 layers)	L4–L16 (13 layers)
3-hop chain	**none**	**none**
Syllogism	L3–L7 (5 layers)	L4–L14 (11 layers)
Negation	L3–L4, L6	L4–L16 (13 layers)
Counterfactual	L4 only	L4–L14, L16, L20

GPT-2: FFN is *never* inactive ($r \geq 0.07$ always). Middle silence = attention off, FFN-only highway.

GPT-2-medium, 1-hop: L7–L13 show **both** attn *and* FFN inactive simultaneously ($r < 0.05$). The residual stream coasts 7 layers **untouched**.

3-hop chain: no silence

The 3-hop chain is the *only* task in GPT-2-medium where every layer shows attn ✓.

This is **counter-intuitive**: the hardest chain task uses the *most* layers, while the easiest (1-hop) uses the fewest — concentrating computation into ≈6 active layers then coasting.

Silence length scales with task

As chain depth increases, the attention-silent zone **shrinks**: the model recruits progressively more middle-layer attention heads to resolve each additional relational hop.

→ Silence length is a proxy for task complexity in chain reasoning.

Slide 4 — Finding 3 & 4: Early Spike Decay + Universal Output Integrator

Finding 3 — Early FFN spike *decreases* with task complexity

The early prominent FFN layer decays monotonically as chain depth increases:

Task	GPT-2 L2 FFN r	GPT-2-med L3 FFN r
1-hop	**1.84**	**1.71**
2-hop	1.41	1.15
3-hop	1.12	0.86
Syllogism	1.43	1.15
Negation	1.42	1.13
Counterfactual	1.34	1.05

Simple tasks front-load more energy into the early FFN (acting as a one-shot lookup). Complex tasks distribute this budget across layers.

Logic tasks (syllogism, negation) cluster near the 2-hop level — consistent with 2-premise retrieval.

Finding 4 — Universal output integrator

The *final* attention layer fires strongly for **every task** in both models:

Task	GPT-2 L11 attn r	GPT-2-med L23 attn r
1-hop	0.89	0.39
2-hop	0.96	0.41
3-hop	**0.99**	**0.42**
Syllogism	0.90	0.38
Negation	0.86	0.41
Counterfactual	0.93	0.38

GPT-2 L11: always ✓, ratio **increases** with hop depth ($0.89 \to 0.99$).

GPT-2-medium L23: always ✓, lower absolute ratio because stream magnitude has grown over 24 layers. GPT-2-medium L22 FFN also shows a consistent "pre-final" spike ($r \approx 0.30$).

Slide 5 — Cross-task Clustering and Summary

Task clustering by silence signature (GPT-2-medium)

Cluster	Tasks
Deep silence (wake $\geq$ L21)	1-hop chain
Long silence (wake L17)	2-hop chain, **negation+de**
Medium silence (wake L15)	Syllogism, **counterfactual**
No silence	3-hop chain (unique)

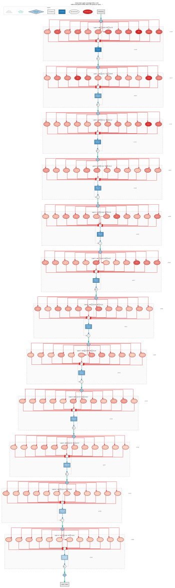
Negation+deduction groups with 2-hop, not with the other logic tasks. Supports the hypothesis that the model routes "No X / All Y / therefore..." through the same 2-premise lookup circuit as a 2-hop chain.

Counterfactual is the least chain-like logic task — its silence is shortest among logic tasks, suggesting additional world-model access.

Summary of findings

Finding	Implication
L0: attn-dominant (GPT-2) vs. FFN-dominant (GPT-2-med)	Different tokenisation strategies; not visible from loss
Attention-silent zones shrink with hop depth	Silence length $\approx$ task complexity proxy
GPT-2-med 1-hop: bilateral dead zone L7–L13	Residual stream coasts 7 layers untouched
Early FFN spike $\downarrow$ as hop depth $\uparrow$	Complex tasks distribute, not concentrate, budget
Final-layer attn r scales with hop depth	Universal output integrator, strength $\propto$ complexity
Negation clusters with 2-hop	Two-premise circuit reuse

GPT-2 (12L×12H) 3-hop Chain $k = 18$ compare_gpt2_task2.png



What to look at

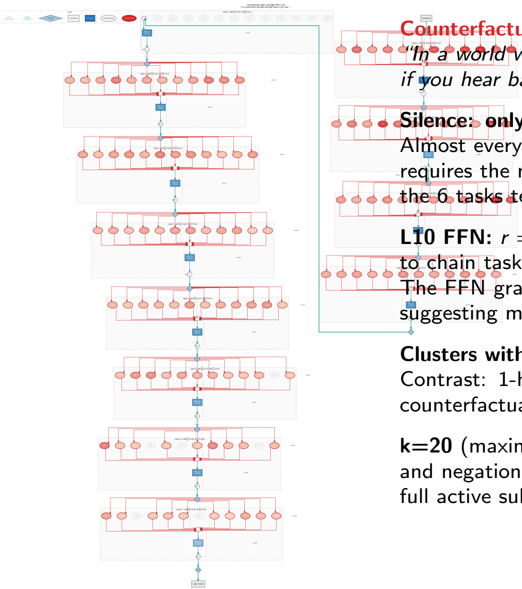
Skip arc colour (left U-shape beside each layer):

- **Thick teal** = block INACTIVE ($r < 0.05$); skip is the dominant path
- Thin dashed = block ACTIVE ($r \geq 0.05$); block transforms the stream

This task: all attention layers ✓
Compare to 1-hop chain where L3–L9 have thick teal U-arcs (attn silent for 7 layers).

L11 final attention: $r = 0.99$ — the highest observed across all tasks. The last-layer integrator works *harder* for deeper chains.

L2 FFN spike: $r = 1.12$ — still above stream magnitude, but lower than 1-hop ($r = 1.84$). Early FFN pre-processing budget shrinks as the task distributes computation deeper.



Counterfactual task

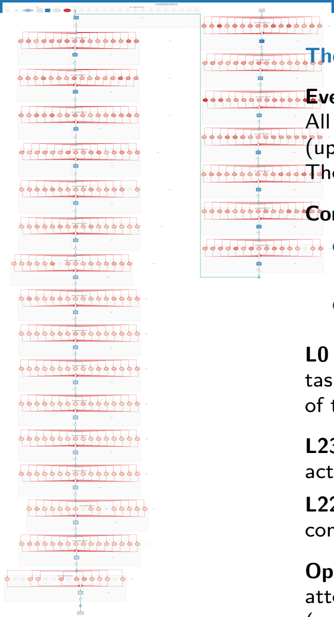
"In a world where cats bark and dogs meow, if you hear barking outside you think it is a"

Silence: only L4 ($r = 0.05$, borderline)
Almost every attention layer fires — this task requires the most distributed reasoning of the 6 tasks tested in GPT-2.

L10 FFN: $r = 0.30$ — notably high compared to chain tasks ($r \approx 0.20$–0.24).
The FFN gradient from L3 upward is steeper, suggesting more sustained semantic refinement.

Clusters with: syllogism (both wake at L4/L5).
Contrast: 1-hop chain wakes attention at L10; counterfactual has attention alive throughout.

k=20 (maximum for GPT-2, same as syllogism and negation) — attribution spread across the full active subgraph, not concentrated.



The anomalous task in GPT-2-medium

Every layer shows attn ✓
All other 5 tasks have long silence zones (up to 17 consecutive attn-inactive layers). The 3-hop chain is categorically different.

Compare with 1-hop (not shown):

- 1-hop: attn silent L4–L20; *both* attn and FFN silent at L7–L13
- 3-hop: **zero silence** anywhere

L0 FFN: $r = 36.3$ (same huge spike as other tasks — the L0 FFN rewrites the token regardless of task type).

L23 attn: $r = 0.42$ — universal output integrator active across all GPT-2-medium tasks.

L22 FFN: $r = 0.29$ — pre-final FFN spike, consistent across all tasks.

Open question: Is the activation of all 24 attention layers a sign of capacity saturation (model cannot solve 3-hop without every layer)?